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The role of fungi in the nutrition of stream invertebrates

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BÄRLOCHER, F., 1985. **The role of fungi in the nutrition of stream invertebrates.** Dead leaves falling into streams are an important food source for many invertebrates. They are generally made more palatable and more nutritious if they are first colonized by aquatic hyphomycetes and other micro-organisms. At least two mechanisms appear to be responsible for this conditioning effect: microbial production (addition of easily digested microbial compounds to the nutritionally poor leaf substrate), and microbial catalysis (conversion of indigestible leaf substances into digestible subunits by microbial enzymes). Different invertebrate species vary in their ability to take advantage of microbial conditioning. This appears to be influenced by their mobility, the range of their food resources and their ability to overcome defense mechanisms of leaf-colonizing micro-organisms.

ADDITIONAL KEY WORDS:—Acquired enzymes – aquatic hyphomycetes – conditioning mechanisms – detritus food webs – microbial catalysis – microbial production – stream invertebrates.

CONTENTS

Introduction 83
Palatability and nutritional value of conditioned leaves 84
Microbial cells as food 85
Other conditioning mechanisms 87
Microbial production versus microbial catalysis 88
Acquired enzymes 89
Two case studies: *Gammarus* Fabr. and *Tipula* L.. 89
Studies with other invertebrates 90
Conclusions 91
Acknowledgements 92
References. 92

INTRODUCTION

Hynes (1963) pointed out that the banks of undisturbed streams are often densely covered with trees and shrubs. This can be expected to have two important effects, especially in small-order streams: incoming light and therefore primary production will be reduced; and the proportion of detritus derived from riparian vegetation will increase. As a result, the biota in many headwater

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streams will depend heavily on allochthonous food sources in the form of leaves, twigs and dissolved organic matter (for a recent review, see Bird & Kaushik, 1981; some important exceptions are discussed by Minshall, 1978). Even though this had been known or at least suspected since the turn of the century (Hynes, 1963), nobody paid much attention to it until Kaushik & Hynes (1971) took a careful look at the fate of autumn-shed leaves in streams. Two of their findings are of special interest.

First, they found that in the early stages of decay the nitrogen and/or protein content of decaying leaves often increases. In laboratory experiments, this increase could only be observed when microbial activity was allowed, and it was generally more pronounced when fungi, rather than bacteria, were the dominant micro-organisms. Since the protein increase was often absolute, that is not simply owing to faster disappearance of other leaf substances, the inescapable conclusion was that fungi and, to a lesser degree bacteria, add protein to the substrate. Kaushik & Hynes (1971) assumed that most of this protein is present in microbial cells. Such microbially colonized leaves are now described as being 'conditioned'.

Secondly, several stream-living detritivores apparently prefer conditioned rather than freshly fallen or sterile leaves (Kaushik & Hynes 1971; Triska, 1970). To explain their observations, Kaushik & Hynes (1971) proposed the following hypothesis: the leaf is basically a substrate for micro-organisms, primarily fungi, whose cells increase the protein content and provide most of the nutrients for leaf-eating invertebrates. Cummins (1973) summarized it as follows: microbially conditioned detritus can be compared to peanut butter crackers in which the protein-rich peanut butter (microbial cells) is embedded in a nutritionally poor cracker. What is the evidence for and against this hypothesis now, some 15 years after it was first proposed?

PALATABILITY AND NUTRITIONAL VALUE OF CONDITIONED LEAVES

The observations on which the hypothesis was based, namely nitrogen increase and improved palatability in decaying detritus, have since been confirmed in many independent studies (for reviews, see Anderson & Sedell, 1979; Bärlocher & Kendrick, 1981; Bird & Kaushik, 1981; Cummins & Klug, 1979). However, some modifications have become necessary. Certain leaves, such as those of alder (Anderson & Grafius, 1975; Nilsson, 1974), have a high protein content and are very palatable even without prior conditioning. At the other extreme, beech leaves (Iverson, 1973) and conifer needles (Bärlocher, Kendrick & Michaelides, 1978; Sedell, Triska & Triska, 1975) require extensive conditioning before they become acceptable.

Regardless of leaf species, there seems to be a definite peak in acceptability; once this is reached, any additional exposure to micro-organisms will lessen the leaf's attractiveness to detritivores. When the effects of microbial activity are being evaluated, care must therefore be taken to distinguish between pre-conditioned, fully conditioned and post-conditioned leaves.

In the study by Kaushik & Hynes (1971) improved palatability was due to an undefined assemblage of terrestrial fungi, whilst in that by Triska (1970) to one of aquatic fungi. It has since been shown that soon after leaves fall into a stream, aquatic hyphomycetes with their tetraradiate or sigmoid conidia, become the

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dominant fungal group (Bärlocher & Kendrik, 1974; Suberkropp & Klug, 1976).

The effects of individual fungal species drastically change the leaf's palatability. Thus, by choosing a suitable fungus, the preference of the amphipod crustacean, *Gammarus pseudolimnaeus* Bousfield, for certain leaf species could be reversed (Bärlocher & Kendrick, 1973a, b). One fungal species, *Tetracladium marchalianum* de Wild., did not improve palatability of the substrate at all. No clear correlation was found between the extent by which a fungus could raise the protein content of a substrate and its palatability. Similar results have been reported by Suberkropp, Arsuffi & Anderson (1983) who worked with caddis-fly larvae (Trichoptera). Again, changes in nitrogen and ATP content caused by a pure culture of one of 10 fungi did not allow any predictions about the leaf's acceptability compared to that of another leaf–fungus combination incubated for the same length of time. In subsequent studies (Arsuffi & Suberkropp, 1984; Suberkropp & Arsuffi, 1984), it became apparent that the length of the conditioning period can profoundly affect palatability. For a given fungus–leaf combination, optimum palatability generally follows the period of greatest growth (as measured by nitrogen and ATP content and activity of degradative enzymes). The length of time that a leaf remains at optimum palatability depends on the fungal species colonizing it. Some fungi had little or no effect on palatability despite their ability to raise the protein content. Unpublished data of T. L. Arsuffi & K. Suberkropp have shown that caddis-fly larvae detect and discriminate between leaf patches colonized by different fungi. Since a leaf in a stream typically supports the growth of five to nine fungal species (Shearer & Lane, 1983), the food of leaf-eating invertebrates will consist of a mosaic of patches which will reach and pass different levels of palatability at different speeds. The passage of a leaf from the pre-conditioned, to the conditioned, to the post-conditioned stage is thus rendered more complex.

Presumably, bacterial colonies on the leaf have a comparable effect. The few observations available indicate that their influence on palatability is less pronounced (Kaushik & Hynes, 1971; Kostalos & Seymour, 1976; Mackay & Kalff, 1973).

As one might expect, the more palatable conditioned leaves are generally used more efficiently than freshly shed leaves. Thus, they are assimilated better or more quickly (Bärlocher, 1982; Golladay, Webster & Benfield, 1983; Grafius & Anderson, 1979; Nilsson, 1974), they allow better survival (Fano, Rossi & Basset, 1982; Kostalos & Seymour, 1976; Rossi & Fano, 1979; Rossi, Fano & Basset, 1983; Rossi & Vitagliano-Tadini, 1978) although no difference was found for oak leaves (Willoughby & Sutcliffe, 1976), and give higher growth rates (Anderson & Grafius, 1975; Rossi & Fano, 1979; Rossi *et al.*, 1983; Rossi & Vitagliano-Tadini, 1978; Sutcliffe, Carrick & Willoughby, 1981; Willoughby & Sutcliffe, 1976).

MICROBIAL CELLS AS FOOD

It has also been demonstrated repeatedly that the microbial component of decaying leaves generally presents a more concentrated source of nutrients than the substrate itself. Thus, various fungal mycelia were assimilated with an average efficiency of 64% (Bärlocher & Kendrick, 1975b), and mixed

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microflora with an efficiency of 72% (Hargrave, 1970; Sedell, 1971; Winterbourn & Davis, 1976). Fungal mycelium loses its cell contents when passing through the gut of *Gammarus pulex* L. whereas the spores of *Mucor* L. and *Aureobasidium* Viala & Boyer remain viable and appear undigested (Willoughby & Earnshaw, 1982). In contrast, the assimilation efficiency for leaves, averaged from a large number of studies, is only 20%, and this includes unconditioned as well as conditioned leaves (Bärlocher & Kendrick, 1975b; Grafius & Anderson, 1979; Hargrave, 1970; McDiffett, 1970; Marchant & Hynes, 1981; Nilsson, 1974; Otto, 1974; Prus, 1978; Sutcliffe *et al.*, 1981; Winterbourn, 1982; Winterbourn & Davis, 1976). This means that microbial food contains two to four times as much nutrient per unit weight as the average leaf.

The results are less clear cut when growth and survival on various microbial foods were measured. May-fly nymphs (a species of *Stenonema*) lost weight on a diet of mycelium of an aquatic fungus (Cummins, 1973). On the other hand, four out of 10 fungi produced higher growth rates in *Gammarus pseudolimnaeus* than did freshly shed, unconditioned leaves (Bärlocher & Kendrick, 1973a, b). Two fungi were toxic, and the remainder fell between these extremes. Kostalos & Seymour (1976) found high to medium survival of *Gammarus minus* Say on variously conditioned leaves, high survival on fungal mycelium and very low survival on unconditioned or sterilized leaves. Growth and survival of *Gammarus pulex* was highest on conditioned leaves and lower on two fungal cultures and unconditioned leaves (Sutcliffe *et al.*, 1981; Willoughby & Sutcliffe, 1976).

In *Asellus aquaticus* L., high growth rates were recorded on conditioned leaves, and on a diet of a *Saprolegnia* species (which is not an important colonizer of leaves in streams) and considerably lower ones for an aquatic hyphomycete and several actinomycetes (Marcus, Sutcliffe & Willoughby, 1978; Willoughby & Marcus, 1979). However, the same isopod species is reported to have gained more weight on several fungal diets than on laboratory-conditioned leaves (Rossi & Fano, 1979; Rossi & Vitagliano-Tadini, 1978). On sterilized leaves all animals died. Of special interest is a comparison between two *Asellus* species which were offered the same food (Rossi & Fano, 1979). In *A. aquaticus*, six out of eight fungi allowed fairly good growth and survival, in *A. coxalis* Dollf. only three of the same eight fungi were adequate. Rossi & Fano (1979) concluded that there is a certain amount of specialization in leaf- and fungus-eating invertebrates not unlike that in herbivores. As a consequence, the usefulness of a fungus to an animal will depend not only on its nutrient content but also on any defensive chemicals it may have evolved and on counter-adaptations of the invertebrate. As mentioned earlier, invertebrates do distinguish between leaves or leaf patches colonized by different fungi even though their protein or nitrogen content may be similar (Arsuffi & Suberkropp, 1984; Suberkropp & Arsuffi, 1984).

Taken together, these results indicate that at least some leaf species allow good survival and reasonably good growth even without prior colonization by aquatic hyphomycetes. Completely sterile leaves have so far proved to be inadequate food. This, of course, may be partly due to the inevitable changes of the substrate caused by sterilization.

Fungi vary considerably as food sources. Some are toxic, some approach or surpass the nutritional qualities of well-conditioned leaves. In experiments reported by Willoughby and colleagues (Marcus *et al.*, 1978; Sutcliffe *et al.*,

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1981; Willoughby & Marcus, 1979; Willoughby & Sutcliffe, 1976) fungal diets were, at best, equivalent to unconditioned leaves (with one exception: *Saprolegnia* allowed growth similar to that on conditioned leaves). A partial explanation for the poor fungal performances in their tests may be based on their technique. They used shaken cultures and they changed the food once a week. In most other studies, mycelium from non-shaken cultures was fed, and the food was replaced every day (Bärlocher & Kendrick, 1973b) or every third day (Rossi & Fano, 1979). In shaken cultures, aquatic hyphomycetes generally form dense, felt-like pellets with many conidia, in non-shaken cultures, hyphae are much more loosely arranged and conidia are rare. After 7 days of exposure to grazing by *Gammarus fossarum* L., the remaining mycelium of two fungal species (*Lemonniera aquatica* de Wild., *Tetracladium marchalianum* de Wild.) release considerably less amino acids and sugars when exposed to the animal's gut fluid (unpublished observations). Thus, the nutritional value of a mycelium can vary considerably with colony age and culture conditions. In addition, presence or absence of toxins may often be more important than nutrient content. Nevertheless, I think there can be little doubt that some aquatic hyphomycetes, grown under the right conditions, represent a relatively concentrated source of food. In addition to protein, their content of lipids and starch-like compounds may also be important, especially for invertebrates close to metamorphosis or reproduction. Their growth on decomposing leaves can therefore appreciably raise the nutritional value of the substrate, as suggested by Kaushik & Hynes (1971).

OTHER CONDITIONING MECHANISMS

It has been estimated that some 2–10% of the weight of conditioned leaves consists of microbial cells, which would deliver a maximum of 25% of the nutrients required by detritivores (Bärlocher & Kendrick, 1981; Rosset, Bärlocher & Oertli, 1982). On beech leaves, these numbers are considerably smaller, less than 1% (Iversen, 1973), yet conditioning still vastly improves their digestibility (Nilsson, 1974). There are several alternative mechanisms.

Freshly shed leaves become conditioned leaves through leaching of soluble substances and through bacterial and fungal activity. In streams, various protozoa, nematodes and algae may also accumulate, although these are not normally present in laboratory-conditioned leaves (Andersen & Sedell, 1979). During this process, the leaves gain nitrogen, not all of it in the form of protein (Odum, Kirk & Zieman, 1979), lignins (Rosset *et al.*, 1982; Suberkropp, Godshalk & Klug, 1976), microbial cells (Kaushik & Hynes, 1971), enzymes (Chamier & Dixon, 1982, 1983; Sinsabaugh, Benfield & Linkens, 1981; Suberkropp & Klug, 1980; Suberkropp *et al.*, 1983) and other secretions (Iversen, 1973). On the other hand, conditioned leaves generally have a lower content of cellulose, hemicelluloses and various soluble organic substances, such as phenols, amino acids and sugars (Rosset *et al.*, 1982; Suberkropp *et al.*, 1976).

Each one of these changes could theoretically contribute to the increased digestibility, though it is unlikely that all of them do. The loss of amino acids and sugars in the first few days of leaching will probably lower, rather than raise, the food value of the substrate. The loss of other substances, such as phenols, in the same period, may have the opposite effect. However, in most

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studies pre-leached leaves have been used and the conditioning effect appears to require an active microbial community.

It is possible that microbial secretions, deposited into the leaf, are just as important as the microbial cells themselves. Iversen (1973) estimated that on beech leaves, over 90% of the nitrogen gained is present in organic substances released by bacteria. A possible mechanism was suggested by Suberkropp *et al.* (1976) who found that a substantial fraction of the nitrogen is fixed in highly refractory complexes, presumably formed from leaf polyphenols and microbial nitrogen compounds (possibly exoenzymes). In conventional analytical methods, these complexes show up as 'lignins'. The relative amount of total leaf nitrogen immobilized in such a way was higher in the slow-decomposing oak (26–36%), with higher initial levels of soluble polyphenols, than in the faster-decomposing hickory. It is at present unknown, if, or to what extent, stream detritivores can use these complexes. At least one of them, *Tipula abdominalis* Say (crane-fly larvae), in which the gut pH reaches values up to 11, seems well adapted to do so (Martin *et al.*, 1980). Under these conditions, polyphenol–protein complexes are no longer stable, and dissociate. In addition, the very active proteases in *Tipula* L. have a high negative charge which presumably prevents their inactivation through adsorption on lignins, humic acids, clay and phenols (Sharma, Martin & Shafer, 1984). High gut pH values have also been recorded in the larvae of one black-fly and three mosquito species (Dadd, 1975; Lacey & Federici, 1979). Martin & Martin (1984) have recently shown that surfactants present in the gut fluids of many insects can prevent the formation of these recalcitrant tannin–protein complexes; it remains to be seen whether they can also dissolve complexes already formed.

In detritus-feeders with gut pH values of between 6 and 9 and less active proteases, for example *Gammarus fossarum* (Bärlocher, 1982), the presence of artefact lignins may simply mean that the leaf's polyphenols are saturated and can no longer interfere with the animal's digestive enzymes.

MICROBIAL PRODUCTION VERSUS MICROBIAL CATALYSIS

The beneficial effects of conditioning mentioned so far are based on the addition of microbial substances, cells or excretions, to the substrate (microbial production, Bärlocher & Kendrick, 1975a). An entirely different mechanism could be based on microbially induced changes of some of the leaf's components which render them more digestible (microbial catalysis). The most likely candidates are structural carbohydrates (cellulose, hemicelluloses, pectin) since they are obviously being attacked by micro-organisms (Chamier & Dixon, 1982, 1983; Rosset *et al.*, 1982; Sinasabaugh *et al.*, 1981; Suberkropp & Klug, 1980; Suberkropp *et al.*, 1983).

There are again at least two ways in which this might benefit invertebrates. Cellulose digestion generally proceeds in steps and is accomplished by a collection of two (bacteria) or three (fungi) major classes of enzymes (Ghose, Montencourt & Eveleigh, 1981). As pointed out by Bärlocher & Kendrick (1975a), several invertebrates are quite efficient at performing the last step in this process; some even show some ability to degrade intermediate breakdown products; however, none of them can use native, crystalline cellulose (Bjarnov, 1972; Monk, 1976, 1977). Their incomplete sets of enzymes will be useful only if

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the animals have access to partly degraded cellulose. *Gammarus pseudolimnaeus* has been shown to recognize and prefer leaves and filter paper partly hydrolysed with hot HCl (Bärlocher & Kendrick, 1975a), and gut fluid of *Gammarus fossarum* releases more reducing sugars from conditioned leaves than from fresh leaves (Bärlocher, 1982), confirming that microbial activity does make polysaccharides more accessible to invertebrates.

In anaerobic environments, respiration is replaced by fermentation, leading to an accumulation of organic acids of low molecular weight. Leaves buried in sediment with restricted oxygen supply presumably undergo these changes and some invertebrates may find such substrata attractive.

ACQUIRED ENZYMES

Another possibility is that microbial carbohydrases, for example cellulases, remain active when ingested by invertebrates. This has recently been demonstrated in a terrestrial isopod (Hassall & Jennings, 1975), a termite (Martin & Martin, 1978), and the freshwater amphipod *Gammarus fossarum* (Bärlocher, 1982). It may well be that the use of such acquired enzymes is the basis for many reports of weak, or sporadic, cellulolytic activity in several other stream invertebrates.

This argument can even be extended to some carnivorous species. When feeding on early instars, they often take in food with little fat or protein, surrounded by an indigestible chitinous shell. The most nutritious portion of such prey may be the digestive tract, packed with food particles. At least to some extent, predators might be considered to feed on prey digestive tracts filled with detritus (Cummins, 1973). The presence of weak cellulases, possibly of microbial origin, may therefore benefit these higher trophic level consumers.

TWO CASE STUDIES: *GAMMARUS* FABR. AND *TIPULA* L.

Gammarus fossarum has a slightly acid (pH 6.0) anterior gut which does not inhibit ingested fungal carbohydrases (Bärlocher, 1982). The posterior part of its gut is alkaline (pH up to 8.5). This allows maximum activity of the animal's proteases which, at this pH level, inhibit or destroy any ingested carbohydrases. Thus, *Gammarus* seems well suited to take advantage of acquired enzymes as well as of microbial proteins.

On the other hand, larvae of *Tipula abdominalis* have a strongly alkaline foregut and midgut (pH up to 11.6) with very high proteolytic activity (Martin *et al.*, 1980). Fungal carbohydrases present on the substrate are inactive at these high pH values (Bärlocher, 1982; Sinsabaugh *et al.*, 1981; Suberkropp & Klug, 1980) and preliminary studies indicate that they are not reactivated in the hindgut where the pH drops to 7.5 (unpublished results). Gut extracts of *T. abdominalis* show no activity towards major plant polysaccharides but can attack α-1,4- and β-1,3-glucans, which, in submerged litter, are concentrated in the associated fungal tissue (they also occur in algae and animals). Chitin, another important, more recalcitrant constituent of fungal cell walls, is not digested. Martin *et al.* (1980) concluded that the digestive system of *T. abdominalis* is geared to efficient protein extraction and digestion. This implies that nitrogen is the limiting factor in its food and that carbohydrate requirements can be

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satisfied by small amounts of easily digested fungal components. In any case, the rather sedentary life-style of *Tipula* presumably keeps energy, and consequently carbohydrate, requirements low.

The high pH of the insect's foregut and midgut may ensure that at least part of the leaf protein, often complexed with tannins and lignins, is set free and becomes exposed to proteases. Presumably, this makes *T. abdominalis* less dependent on more easily digested microbial proteins. Of course, it should still do better on conditioned leaves with their higher protein content, and this has indeed been shown (Cummins & Klug, 1979).

The digestion strategy adopted by *G. fossarum* will be most successful if well-conditioned leaves are plentiful. If this is not the case, it will be viable only on highly mobile animals that can search out the few well-conditioned leaves, or in animals with alternative sources of food. Conversely, the evolution of very efficient nitrogen digestion in *T. abdominalis*, a slow-moving, obligate leaf shredder (Cummins & Klug, 1979), could indicate that in streams pre- or post-conditioned leaves predominate.

This was proven to be the case in a recent study of a hardwater stream in the Swiss Jura (Bärlocher, 1983). The average protein content of randomly collected leaves declined by approximately 50% between November and April. Digestibility of these leaves (characterized by the release of reducing substances and amino acids from ground-up leaves exposed to gut extracts of *Gammarus fossarum*) also declined in the same period. Average values were 10–30% of the maximum reported for laboratory-conditioned leaves. However, variability was great and some of the stream leaves were comparable to laboratory leaves. For a mobile, and facultative, shredder, these higher values are probably more relevant than the mean value. In *G. fossarum*, the major detritus-feeder in that stream, the contribution of leaves to its diet dropped from 85% in fall when most leaves were conditioned, to approximately 30% in spring when most leaves carried little microbial biomass, and had a high lignin and a low cellulose content (Rosset *et al.*, 1982). A less mobile, obligate leaf-feeder would have to counterbalance this steady decline in the quality of its food by using it more efficiently. An appropriate adaptation would appear to be a high gut pH: 60–90% more protein was soluble at a pH of 11 than at a pH of 8 (corresponding to the pH of *Tipula* and *Gammarus* guts, respectively).

STUDIES WITH OTHER INVERTEBRATES

At present it is not known whether the digestive and behavioural adaptations of *Gammarus* or of *Tipula* are more representative of a typical leaf shredder. Most detritus-feeders are quite flexible and do not depend completely on any one food item (Anderson & Sedell, 1979; Cummins & Klug, 1979). This allows many of them to discriminate against less well-conditioned leaves, as illustrated by the repeatedly observed delay in the colonization of leaf packs in streams. The lag period is generally longer and more distinct in nutritionally poor leaves, provided that sufficient alternative foods are available (Haeckel, Meijering & Rusetzki, 1973). The precise nature of the attraction of conditioned leaves is complex and species specific. Neither protein content of a leaf–fungus combination, nor its carbohydrase activities, gives any reliable clue to its palatability as compared with that of another combination (Arsuffi &

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Suberkropp, 1984; Suberkropp *et al.*, 1983). Presumably, specific organoleptic substances are involved: repellants, attractants and toxins, which may override purely nutritional considerations.

Even if the role of these secondary substances is neglected for the moment, remarkably little is known about which substances are actually important for the nutrition of detritus-feeders. Larvae of two stone-flies (Plecoptera) and three caddis-flies (Trichoptera) had a neutral or slightly alkaline gut pH (Martin *et al.*, 1981, 1982), very active proteases, negligible activity towards structural polysaccharides of higher plants and could digest α-1,4- and β-1,3-glucans (present in fungi, algae and animals). The pH values would allow activity of ingested carbohydrases, provided they are not inactivated by the proteases.

The obligate shredder *Zelandopsyche ingens* Tillyard (Trichoptera) from New Zealand digests much of the microbial biomass on the dead leaves (Winterbourn & Davis, 1976). It may also profit from cellulose digestion by bacteria residing in the foregut (Winterbourn, 1982). The potential contribution of enzymes acquired with the substrate and the pH profile of the digestive tract were not studied.

High pH values have been recorded in the larvae of one black-fly and three mosquito species (Dadd, 1975; Lacey & Federici, 1979). They all belong to the so-called collectors (Cummins & Klug, 1979), which ingest fine particulate organic matter. A variable proportion of this substrate consists of finely shredded leaves and its food value is quite low, especially after repeated ingestion and egestion by detritus-feeders. A significant portion of the food eaten by collectors may, therefore, be of even poorer quality than that of shredders, again favouring more exhaustive nitrogen extraction.

CONCLUSIONS

On evaluation of the hypothesis by Kaushik & Hynes (1971), I think there can be little doubt that the simple and elegant mechanism they proposed, the improvement of a poor substrate by the addition of valuable nutrients, in cells or as acellular secretions, can contribute to the conditioning effect. I think it has also become obvious that this is not the only mechanism, nor is it always effective. The leaf itself is not an inert substrate but changes under microbial influence. Conversely, fungi and bacteria are not simply passive containers filled with valuable food. They are living organisms, and as such, can be expected to evolve adaptations to avoid losing their protoplasts and their food sources (leaves) to foraging invertebrates. These countermeasures may involve 'hiding' from invertebrates, for example by forming colonies that cannot be detected by the animal; deterring potential consumers by producing repellants; or the ability to pass unharmed through the digestive tract of detritivores (Bärlocher, 1979). These factors are likely to complicate the issue.

I believe that a reasonably clear perception of the main mechanisms involved in conditioning is available, but the testing of their actual involvement in natural streams has only just begun. To find out more, studies on the digestive physiology of selected invertebrates, combined with observations on their feeding behaviour and mobility, and an assessment of the availability and condition of the various food resources will be needed. Eventually, new generalizations may emerge concerning the mechanisms that allow aquatic

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detritus-feeders to meet their nutritional requirements. The involvement of aquatic hyphomycetes and other micro-organisms will be a crucial factor in these generalizations.

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